that has taken place is in connection with the removal of main and side rods, though removal of main rod simply because of disconnecting valve stem and covering ports with valve has become so rare that it seems hardly necessary to make it a subject of discussion here.

The established practice in regard to disordered side rods has been: *In case of failure necessitating removal of side rod or section of side rod the corresponding side rod or section of side rod on opposite side should also be removed.* And, *in case of failure of front section of side rod on locomotives having three pair, or of intermediate section on those having four pair of connected wheels, remove all side rods on both sides and run locomotive with only main rods connected.* Another, *in case main crank pin breaks off close to wheel center, remove all side rods on both sides and run locomotive with but one main rod connected.*

The first proposition is undoubtedly the correct method when the side rod involved is either a front or back section on any type of locomotive with four pair of connected wheels (Mallet types with two groups included), and back section of types with three pair connected (Mallets included); but when a disorder necessitates the removal of all side rods on the disordered side of a locomotive, there are circumstances in which it may be much better practice to allow the side rods to remain connected on opposite side. The removal of side rods on both sides of a locomotive reduces its available adhesion in proportion to the number of drivers remaining connected, that is, if side rods were removed from both sides of a 4–4–0 or a 4–4–2 type the adhesion would be reduced about one-half; if removed from both sides of a 2–8–0 type it would be reduced to one-quarter. With adhesion reduced to such low points locomotives could not satisfactorily handle trains owing to excessive slippage.

It has been satisfactorily demonstrated that where a disorder necessitates removal of all side rods on one side, under ordinary circumstances the best method of procedure is to allow side rods on the opposite side to remain connected, and that such practice results in saving in time required to disconnect the disordered parts, gives more satisfactory road performance and enables a locomotive to handle a heavier train than if all side rods are removed, though it must be remembered proper precautions must be taken when this method is followed. In all cases when this method is followed *the main rod must be removed on the disordered side, or at least the engine on that side must be put out of commission by disconnecting the valve stem and covering ports with valve.*

The reason for putting engine on disordered side out of commission is so that the drivers will not slip when the crank pins on connected side are on the center, straining or buckling the side rods. This is the only point where a straining action on side rods would have a tendency

gets its steam supply, because the water acquires momentum owing to the steam's imparted velocity, overcoming the pressure in the boiler. The steam, in forcing water through an injector into a boiler, gives up its heat and performs actual mechanical work as truly as though it acted on a piston moving a pump plunger with it. The heat of the steam is not lost, excepting that that represents the mechanical equivalent of work performed in forcing water to the boiler, for it is stored in the feed water, raising its temperature.

Locomotive injectors are operated by steam from the locomotive's boiler. Each gallon of water fed to the boiler requires about two cubic feet of steam to operate the injector and owing to the steam's condensing its volume when returned to the boiler is only one-sixth of a cubic foot, and so if the boiler does not generate steam as fast as used by the injector in addition to that used by the engine and other steam-using apparatus about the locomotive, a fall of steam pressure results, which varies with volume of steam space in the boiler and conditions under which the locomotive is operating.

Injectors used on locomotives are of two classes, the *lifting* and the *non-lifting*. The lifting injector is capable of lifting water from a lower level than its own, while the non-lifting must be located lower than the water level. The principal parts of both lifting and non-lifting injectors are a *steam nozzle* from which the steam issues in the form of a jet when the injector is in action; a *combining tube* where the steam and water meet, steam condensed and its energy transmitted to the water; a *delivery tube* in which the water attains its maximum velocity; a *check valve* to prevent flow of water from the boiler when the injector is not in action; *hand operative valves* for the admission of steam and water, and an *overflow valve* that permits circulation of steam through the injector when in the act of priming.

OPERATING INJECTORS.

Engineers taking charge of a locomotive should satisfy themselves that both injectors are in good order, and the best way to keep them in that condition is to operate both—*i.e.*, have each injector perform an equal amount of service.

To prime a lifting injector steam is admitted to the steam nozzle, which forms a current in the direction of the combining tube. This current induces the air in the supply pipe above the water to flow in the same direction, creating a partial vacuum in the supply pipe; the atmospheric pressure on the water supply forces the water up to fill this vacuum, thus causing a constant flow to the injector while it is in action. After an injector is primed its proper action is obtained by increasing the steam supply and closing the overflow valve.

In shutting off an injector the lever should be operated easily, so

Q. How should a test be conducted in order to locate internal blows?
Q. What important duty should engineers perform when accidents or breakage of locomotives necessitate removal of parts?
Q. What should be done if an intermediate or middle section of side rod on a 2-8-0 or 2-8-2 type should break?
Q. What should be done if a front or back tire on a 4-4-0 type should break?
Q. What should be done if either a front or main tire on a 4-4-2 type should break?

PART X.

Q. How should a Sellers Self-acting Injector be operated?
Q. How should a Sellers Non-lifting Injector be operated?
Q. How should a Nathan Non-lifting Injector be operated?
Q. What should be done if the water inlet valve of Simplex Injector leaks, preventing the prompt lifting of feed water?
Q. How should a Nathan Non-lifting Injector be operated?
Q. How should an "Ohio" Lifting Injector be operated?
Q. How should an "Ohio" Lifting Injector be operated to use as a heater?
Q. How should a Chicago Standard "R" Non-lifting Injector be operated?
Q. How should a Chicago Standard "R" Non-lifting Injector be operated to use as a heater?

PART XII.

Q. What law fixes the responsibility for the general construction, safe working pressure, inspection and testing of locomotive boilers and their appurtenances?
Q. What is the lowest factor of safety for locomotive boilers permitted by law?
Q. What is the tensile strength of steel boiler plate? Tensile strength of wrought iron boiler plate?
Q. How often must stay bolts be tested?
Q. How often must steam gauges be tested?
Q. Where should the metal badge plate, showing the allowed steam pressure, be attached to the boiler?
Q. How often should the spindles of all gauge cocks and glass water gauge cocks be removed and cleaned of scale and sediment?
Q. Is it permissible for gauge cocks to be in such bad order that they cannot be manipulated without the aid of a wrench or other tool?
Q. In case of accident what does the law require?

the front end of the radius rod from the valve stem crosshead guide, using for this purpose a wire or chain (the fire door chain if no other is at hand). Secure the valve to cover the ports, and if necessary remove the main rod and block the crosshead at the back end of the guide.

NOTE.—If the engine has cylinder relief valves and the main rod and piston on the disabled side are in condition to run, the main rod may be left up when the valve is secured to cover the ports. In that event remove the relief valves from the cylinder heads to relieve compression and also permit of lubricating the cylinder.

If the lap and lead lever, lap and lead lever connector, or crosshead arm break; with inside admission valves and relief valves in the cylinder heads an engine can be blocked, as shown in Fig. 59. Disconnect the radius rod from the lap and lead lever and suspend it clear of the latter, as shown. Secure the valve to cover the ports. Tie the lower end of the lap and lead lever ahead by means of a rope secured to the cylinder cocks. Take out the relief valves. The engine can then be run with one side.

With outside admission valves under the same conditions as illustrated in Fig. 59, the main rod could be left up and the engine disconnected in the same way, except that the lap and lead lever would have to be taken down in order to clear the radius rod.

BAKER GEAR.

With the Baker Gear there are two ways for blocking the gear and valve in case of breakdown. The valve stem crosshead is provided with a set-screw so that the valve can be blocked central over the ports by clamping the valve stem crosshead to its guide. This is done in case the breakage of the gear or engine disables one side.

The other way of blocking is by bolting the lower arm of the bell-crank fast to the side of the frame. After the valves have been set, the reverse lever is put in midgear and a hole drilled through frame and bell-crank arm. Any bolt that will go through the hole can be used. (See Figs. 52 and 53, Baker Gear, in Part VII.)

In the latest designs of Baker Gear two holes are drilled into the gear frame and a clamp is provided to encircle the bell-crank and hold it stationary. This provision makes it practicable to get the lap and lead movement when the following parts fail: Eccentric crank, eccentric rod, gear connecting rod, radius bar, reverse yoke, short reach rod and horizontal arm of bell-crank. In case any of these parts break bolt the bell-crank arm fast and remove the eccentric rod. With the gear bolted in this manner the valve will get the lap and lead movement and a port opening equal to the lead for all cut-offs, obtaining the same advantages as outlined in second paragraph, Breakdowns, Walschaerts Gear.

placed across frame in front of cylinder saddle. Run carefully with a locomotive blocked in this manner.

Broken Front Axle.—4–4–0 type. When a front axle breaks outside the driving box, disconnect the main rod on that side, remove the side rod, disconnect the valve rod and clamp the valve stem with valve in mid-position. Run carefully without train.

Broken Back Axle.—4–4–0 type. Remove both side rods, block up the back pair of wheels the same as for broken tire except removing the oil cellars. Proceed with as much train as the locomotive will haul with front pair of drivers.

Broken Front Axle.—4–4–2 type. Remove both side rods, jack up end of axle until it is in its normal position, remove cellar and block between the journal and pedestal brace, also block between top of frame and spring saddle so as to take the weight off of box and place it on the frame.

Broken Main Axle.—4–4–2 type. Remove both side rods and the main rod on the disabled side, drop the main wheel on the disordered side out of the way and block the same as for the front axle. Disconnect the valve stem and cover the ports on the disordered side.

Broken Front Axle.—All types with three or more pair of drivers. When the front driving axle breaks outside the box it is necessary to remove the front sections of both side rods. Remove the driving wheel, jack up the broken end of axle until it is in its normal position blocking between the journal and the pedestal brace, and block between the top of the frame and the spring saddle in order to relieve the box of weight. If the following pair or pairs of drivers have plain tires, great care must be exercised in moving around curves in order that the wheels with plain tires may not drop off of the rails.

Broken Second Axle.—All types with four or more pair of drivers. If the axle breaks outside the box it is necessary to remove all side rods. Jack up broken end and block up between it and pedestal brace same as for broken axle 4–4–2 type and block between top of driving box and spring saddle to relieve box of its load.

Broken Main Axle.—All types with four or more pair of drivers. If no other damage should be done remove all side rods and main rod on disordered side. Block same as for second axle. If main wheels have blind tires and no collars on axles it will be necessary to swing main wheel to clear the rail on opposite side. In this case both main drivers will be out of service and the engine must be towed.

Broken Main Driving Spring or Hanger.—4–4–0 type. Run the main driver of the disordered side up on a block so as to relieve the back spring, pry up the front end of the equalizer into a level position and block it there, run main driver off of the block, and the back driver up on one and block on top of the main box under the frame. After running the engine off of the block, slacken down the main

wedge to prevent box sticking if it should run hot and keep wedge and shoes well oiled.

Broken Back Driving Spring or Hanger.—4-4-0 type. With an engine having overhung spring rigging, run the back wheel up on a block to take strain off main spring and pry the back end of equalizer level and block it, run back wheel off block and main wheel up on one and place a block on top of the back box under the frame levelling the engine, slacken the back wedge and keep driving box well oiled.

Broken Equalizer.—4-4-0 type. Run back driver up on a block so that a piece of iron or large nuts can be placed on top of the main box under frame bar. Then run the back driver off the block and main driver up on one, and block between back driving box and frame. The broken equalizer and loose spring rigging should be removed to prevent further damage and boxes oiled freely, as they are liable to run hot.

Broken Main Driving Spring.—4-4-2 type. On locomotives having overhung spring rigging, run main driver up on a block, to relieve the springs, pry up back end of front equalizer to its normal position and place a block between back end of equalizer and bottom rail of frame. Pry down the back end of back equalizer and block between top rail of frame and back equalizer. Run main driver off of block and front driver up on one, block between top of main box and top rail of frame. Remove broken or loose parts to prevent further damage. This method permits of using front and trailer springs.

Broken Front Spring or Hanger.—4-4-2 type. Run the front driver up on a block to relieve spring rigging, pry down on back end of front equalizer until it is in normal position, and block between the front end of this equalizer and the bottom rail of the frame. Run the front driver off of block and the main driver up on one and block between the top of the front box and frame, thus permitting the front driver to carry its share of the load.

NOTE (Broken Tires).—In all cases of broken tires the objective point is to raise the wheel center a height equal to the thickness of the tire, thus levelling the axle, blocking in such manner as to relieve the driving box over journal of disordered wheel, redistributing the weight over the other boxes. When the tire of main wheel is broken it will be necessary to put engine out of commission on disordered side, removing main and side rods if damaged at time of tire breakage. When the tires of back wheels are broken and there are no flanges on the tires of the pair ahead it will be necessary to hold back end of locomotive over toward the side having good tire, which may be done by fastening a heavy chain from the tail piece of frame on disordered side to front end of opposite side of tender. This method will guide back end of locomotive and keep it on the rail, though it must be run carefully on curves and over frogs and switches. Tire breakage hap-